

REVIEW ARTICLE

Communications in Space–Air–Ground Integrated Networks: An Overview

Kai Yang¹, Yichen Wang¹, Xiaozheng Gao^{1*}, Chenrui Shi², Yuting Huang¹, Hang Yuan¹, and Minwei Shi³

¹School of Information and Electronics, Beijing Institute of Technology, Beijing 100081, China. ²School of Information and Communication Engineering, Communication University of China, Beijing 100024, China.

³School of Cyberspace Science and Technology, Beijing Institute of Technology, Beijing 100081, China.

*Address correspondence to: gaoxiaozheng@bit.edu.cn

As an emerging vertical heterogeneous network architecture that integrates the satellite-based space networks, air networks, and traditional ground networks, space–air–ground integrated network (SAGIN) is developed to realize the Internet of everything, global coverage, and ubiquitous intelligent communications. However, SAGIN also faces quite a few challenges due to its unique characteristics, such as highly complex network architecture, highly dynamic node topology, time-varying communication channels, and restricted resources. In this paper, we first introduce the architecture and benefits of SAGIN, and then present the faced challenges. Next, we discuss some key technologies in SAGIN. In particular, we discuss the flexible access in the realization of a dynamic communication architecture, explore some efficient resource scheduling methods to avoid wastage of resources, and investigate the secure communications, such as covert communications, physical layer security, and anti-jamming communications, in SAGIN. Finally, the potential future directions are discussed.

Introduction

In recent years, with the successful commercialization of the fifth-generation (5G) mobile communication networks, smart devices, such as smart phones and intelligent robots, have been rapidly developed, and there is an exponential explosion in network traffic [1,2]. Therefore, seamless, ubiquitous, and global connectivity has been an exigent requirement in future communications. However, the current traditional ground communication network still fails to meet the needs of providing seamless and high-quality coverage of remote and rural areas [3]. Also, due to the inherent disadvantages, like insufficiency in withstanding natural disasters, utilizing the traditional ground communication network alone cannot achieve reliable communication quality requirements on a global scale [4]. As a result, it is necessary to develop a network framework that aligns with the multifaceted requirements of the expeditious evolution of the information society for diverse services and applications.

An emerging vertical heterogeneous network architecture, called space–air–ground integrated network (SAGIN), is developed to compensate for the shortcomings of the ground network, aimed at providing better services in realizing the “4A” (Anytime, Anywhere, Anyone, and Anything) communication scenarios [5]. With the advantages of modern information technologies, SAGIN integrates the satellite-based space networks, air networks, and traditional ground networks deeply in terms of system compositions, key technologies, and applications [2]. Interconnecting space, air, and ground network segments, the

ideal SAGIN can provide network services covering the whole earth with high-quality communications, by making full use of the functions of various network dimensions and eliminating the obstacles of information sharing among different network segments.

The architecture of SAGIN is shown in Fig. 1. The 3 network segments that integrate SAGIN, namely, space, air, and ground, can either work cooperatively or perform independently. A space network is constituted by spaceships, satellites, constellations, and their corresponding ground infrastructures, such as fixed ground stations, mobile stations, user terminals, and network operation control centers. The satellites and constellations in different orbits possess distinct features [6]. In particular, according to their channel bandwidth, the satellites can be categorized into narrow-band and broadband networks. Also, according to the altitude, the satellite networks can be classified into 3 categories: geostationary earth orbit (GEO), medium earth orbit (MEO), and low earth orbit (LEO) satellites [7]. Different from the ground communications, satellite communications can evolve 2-dimensional “population coverage” at the surface level into 3-dimensional “global space coverage”. The space network can provide a global coverage on the earth, but the long propagation latency and the limited capacity should be concerned [8]. The air network mainly makes use of unmanned aerial vehicles (UAVs), airships, and balloons to constitute the high- and low-altitude platforms (HAPs and LAPs), using aircraft to support efficient communications [6,9]. These platforms have the ability to offer lower cost, more

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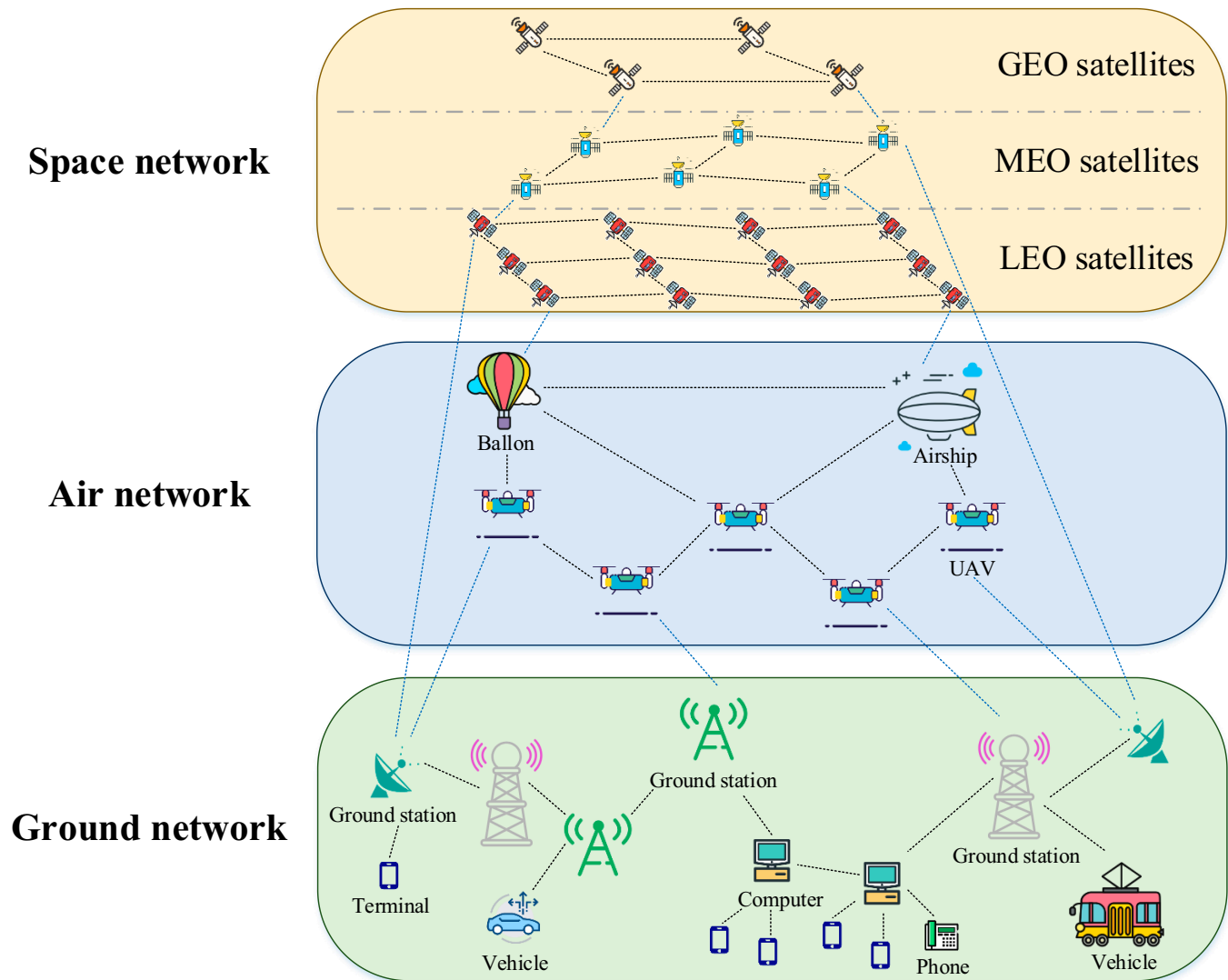


Fig. 1. An architecture for space-air-ground integrated networks.

flexible deployment, and a wider range of services to complement the ground networks [10,11]. The ground networks, such as cellular networks, have become an indispensable part in our social daily life, offering low-latency and high-data-rate ground communication services for most urban spots. In particular, cellular networks have successfully evolved to 5G wireless networks, and now are evolving to the forthcoming sixth-generation (6G) mobile networks to support diverse distinct application services [6].

Overall, we can foresee the opportunity that the SAGIN holds great potential in achieving multi-dimensional ubiquitous coverage and information interaction, and it enhances important support for some future applications, such as emergency rescue, intelligent transportation, and ecological remote sensing [2]. For instance, the integration of the communication, guidance, and sensing capabilities of the satellites in SAGIN can contribute to the intelligent, reliable, and integrated command of emergency rescue, improving the ability of earthquake prevention, disaster reduction, and catastrophe relief. Also, the deployment of UAVs in the air networks provides the opportunity to improve communication quality in the areas with complicated terrain or weak signal strength,

such as mountainous areas and high-altitude areas. For instance, the acquisition of road features in the abovementioned areas can be facilitated by the deployment of UAVs, which enhances reliable location services for both aerial and vehicular terminals [12]. In addition, the dense and scattered ground networks are able to provision large data rate access for massive users. Through the deployment of the 3-layer heterogeneous network, SAGIN has the capacity to provide pervasive coverage across all the areas without any blind spots. This sophisticated infrastructure enables seamless and precise ecological monitoring, extending its functionality to intricate and remote environments. Therefore, SAGIN-based 6G is expected to realize the broadband and Internet of Things (IoT) communications globally [12,13], and the integration of space, air, and ground is an important way for future networks to achieve seamless global coverage [14]. Nevertheless, it is noteworthy that such high-quality idealized services also come up with strict requirements for the architecture design of SAGIN accordingly, making it more challenging than traditional ground networks [9].

As shown in Fig. 2, the remainder of the overview is structured as follows. In the “Challenges” section, we analyze the

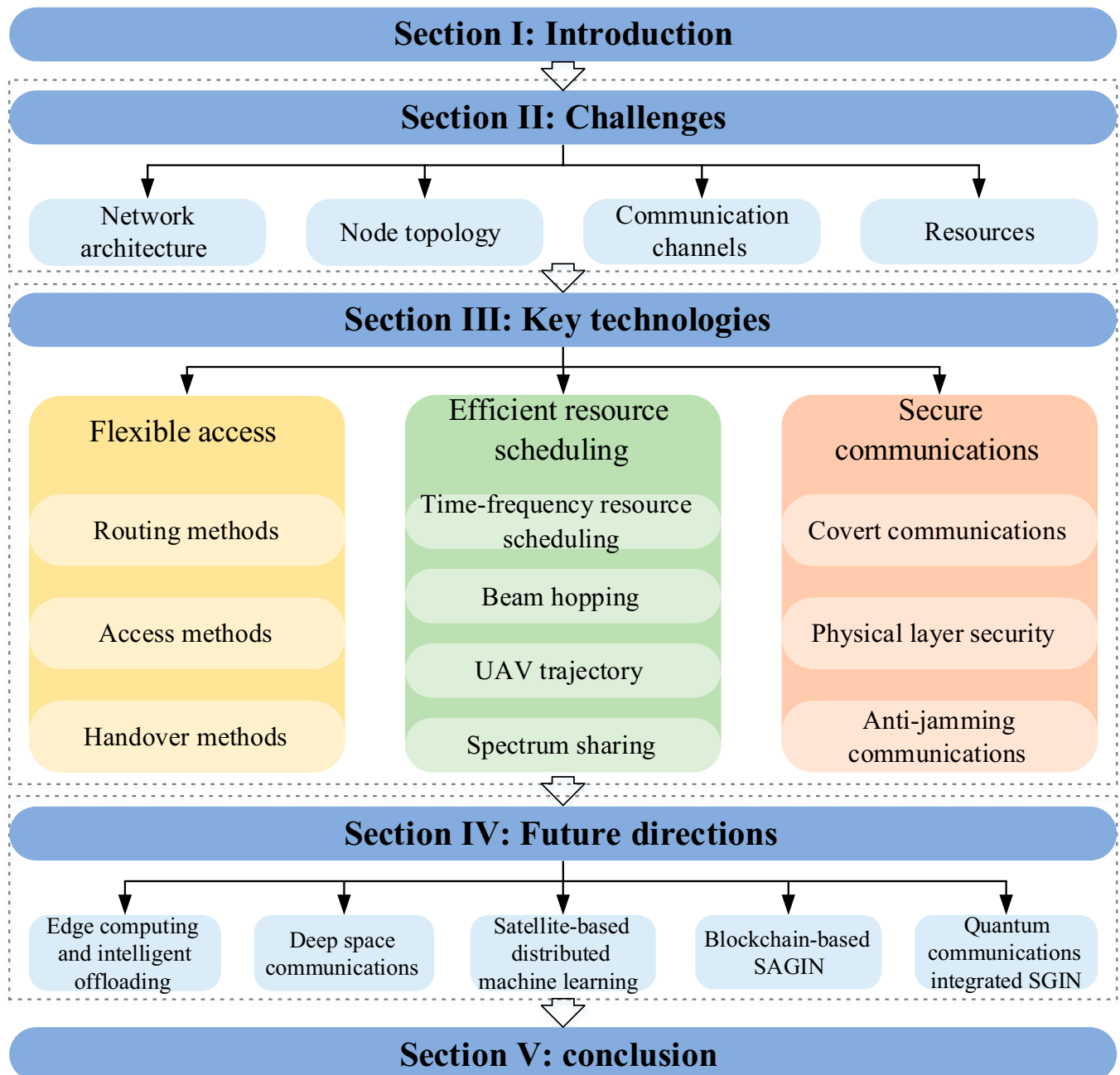


Fig. 2. Structure of the overview.

challenges faced by SAGIN. Then, we detail the key technologies in the “Key Technologies” section. Finally, we point out some future directions for SAGIN in the “Future Directions” section. The abbreviations used in the overview are summarized in Table.

Challenges

Although there are many benefits in SAGIN, the current SAGIN is still in its nascent phase and faces numerous challenges [15]. In this section, we analyze the main challenges faced by SAGIN, from the aspects of the network architecture, node topology, communication channels, and resources.

Network architecture

The SAGIN, whose subnetworks have diverse network environments and distinct structures, spans from the surface of the earth to the space orbit, showing a highly complex network architecture and multi-layered 3-dimensional characteristics.

Compared with traditional ground networks, the integration, security, and management of SAGIN is much more challenging due to the highly complicated network architecture [16]. First, the operations in such a complicated network environment should consider the network integration rigorously by effectively integrating and coordinating the infrastructures of the space, air, and ground network segments to provide seamless communication services [6]. Second, due to the inherent nature of its multi-dimensional heterogeneous architecture,

Table. Abbreviations used in this overview

Abbreviations	Full name	Abbreviations	Full name
5G	Fifth-generation	RMA	Random multiple access
SAGIN	Space–air–ground integrated network	TDMA	Time-division multiple access
GEO	Geostationary earth orbit	MIMO	Multiple-input multiple-output
MEO	Medium earth orbit	AI	Artificial intelligence
LEO	Low earth orbit	ISL	Inter-satellite link
UAV	Unmanned aerial vehicle	DCA	Dynamic channel allocation
HAP	High-altitude platform	FDMA	Frequency-division multiple access
LAP	Low-altitude platform	MF-TDMA	Multi-frequency time-division multiple access
6G	Sixth-generation	SNR	Signal-to-noise ratio
IoT	Internet of Things	DSSS	Direct sequence spread spectrum
ATM	Asynchronous transfer mode	AWGN	Additive white Gaussian noise
IP	Internet protocol	SRL	Square root law
QoS	Quality of service	AN	Artificial noise
OMA	Orthogonal multiple access	IRS	Intelligent reflecting surface
NOMA	Non-orthogonal multiple access	RF	Radio frequency

disparate network segments introduce various security requirements, significantly increasing the security vulnerabilities in the SAGIN framework [17]. Therefore, it would be a tough challenge to implement reinforced security measures to safeguard the system from malicious attacks, and ensure the privacy of users [18]. Third, the demand on data management in SAGIN is much more strict than that in the ground network due to its ubiquitous coverage and diverse business types; thus, it is imperative to employ efficient data management mechanisms to guarantee the performance of real-time applications and decision-making processes. Besides, in SAGIN, it should be flexible to hand over among various network segments as required so that the end-user terminals can fully leverage the benefits of widespread coverage offered by nonground networks as well as low-latency advantages of ground networks [2]. However, since the satellites generally move faster than the ground, the handover issue among multi-layer networks faces huge challenges, and a special mobility management scheme is needed. Finally, it is of great significance that the SAGIN should continuously integrate resources among space, air, and ground, and needs to be equipped with strong autonomous management ability as well as fast response ability in emergencies. In summary, designing an effective architecture for SAGIN is of special importance.

Node topology

Composed of different levels of nodes, such as diverse satellites, HAPs, LAPs, and ground equipments, SAGIN is formed by the establishment of various communication links, which include inter-satellite links (ISLs), satellite-to-ground links, and air-to-ground links. The high mobility of many nodes determines that the communication links between different nodes will be frequently switched, which constitutes the dynamic node topology of SAGIN.

The highly dynamic network topology of SAGIN gives rise to substantial challenges for designers in many aspects, such as

robust fault tolerance, real-time information exchange, intricate routing optimization, and meticulous data synchronization. For instance, due to the highly dynamic features of the nodes, the communications in SAGIN face significant Doppler frequency offset, which causes a large outage rate and high bit error ratio [19]. Besides, the rapidly changing network topology may lead to increased real-time requirements, especially for the applications that demand low latency and high reliability. At the same time, the high-speed changing topology of SAGIN will generate changes of network links, making the transmission of data through fixed flows difficult. In addition, as the physical layer channels change with the communication nodes in SAGIN, the upper-layer decisions including flow control and routing strategies are affected, which in turn lead to the fact that the latency, data transmission quality, and efficiency of the whole system are affected [16]. In summary, there is a necessity to develop new methods related to distinct layers for network resources management and optimization, and adopt appropriate data synchronization mechanisms to ensure data consistency among various nodes.

Communication channels

The communication channels in SAGIN manifest significant variations, and the channel parameters of SAGIN exhibit time-varying characteristics, presenting a notable distinction from the extensively investigated ground communication channels.

Particularly, the rapid changes in channel parameters pose a series of issues, such as instantaneous signal attenuation, uneven spectrum utilization, and dynamic variations in multipath effects, which in turn affects the reliability and stability of communications, reduces spectrum efficiency, and results in a decline in the quality of data transmission, especially in mobile and complex environments [1]. Hence, it is urgent to develop a well-adaptable channel model according to the characteristics of the communication channels in SAGIN. However,

it differentiates from the conventional assumption in the majority of channel models, where the distribution across time, frequency, and space domains is considered to be stationary [6,20]. In SAGIN, the channel conditions fluctuate with many factors, such as time, geographical location, weather conditions, and flight altitude. Besides, the communications related to the widespread satellites, UAVs, ground terminals, and other space platforms have completely different channel modes. Overall, the channels in SAGIN generally exhibit statistically rapid variations, which brings huge trouble for the efficient and accurate channel modeling. The current satellite channel models are mainly proposed for suburbs and open spots, while the models in the scenarios with complicated scattering environments, such as low-orbit dense constellation service cities, are still in the argumentation stage and require further development. In summary, due to the time-varying characteristics of channels, there is a lack of propagation channel models for SAGIN that have been extensively tested and verified.

Resources

Due to the constraints of volume, weight, and other conditions of the equipments, such as satellites and UAVs, the computing resources and energy resources in SAGIN are often restricted. Besides, the resources for the production of the massive infrastructure in SAGIN are also limited.

The limited and unbalanced resources in SAGIN present considerable difficulties in effectively optimizing the resource utilization and considering a balance between performance and resources. Specifically, the rise of computing-intensive applications, such as unmanned driving, virtual reality games, and remote intelligent robots, has brought about an explosive growth in the amount of information data and computing volume, while the limited computing resources of the single node and the unbalanced distribution of computing resources of the multiple nodes in SAGIN restrict the demand of users for lower latency and higher quality service experience [21]. Therefore, it is urgent to fully improve the resource utilization of SAGIN by scheduling time, space, and frequency wireless resources, computing resources, storage resources, and interface resources in SAGIN so as to realize the vision of ubiquitous high-precision and fast computing capabilities in the future 6G. From the perspective of energy constraints, the following is an example of the limited on-board resources of satellites. The power resources of satellites are limited, and the power consumption of the equipments of the satellites is largely limited by the thermal radiation capability of the satellites [22]. However, due to the limited weight and size of the satellites, the size of the thermal radiator cannot be arbitrarily increased, and the thermal radiation capacity of the satellites is limited [16]. With the rapid development of the requirements of satellite communications for higher data rate, lower latency, and more connections, the contradiction between limited load resources and increasing demands on transmission power will be further intensified [16]. Thus, we should pay more attention to the energy efficiency issues and carefully design all the elements in SAGIN as well as its underlying backhaul network so that the overall power consumption can be as minimized as possible [1]. For constraints on resources, SAGIN must make the performance trade-off and meet various demands of the equipments in SAGIN. Besides, considering the significant consumption of resources and energy as well as the emission of pollutants brought by the launches of satellite-carrying rockets, the production of rocket's

fuel, and the manufacture of components of satellites, along with the high environmental costs associated with the energy sources needed to power the UAVs, such as batteries and fossil fuels, it is crucial to address the notable environmental impacts and sustainability challenges in the deployment of the SAGIN infrastructure. Efforts to integrate the renewable energy, use eco-friendly materials for manufacturing the components of satellites and UAVs, and design longer-lasting satellites as well as UAVs can help maximize the benefits of the SAGIN while minimizing the environmental problem.

Key Technologies

In current SAGIN, key technological pillars include flexible access, efficient resource scheduling, and secure communications. These pillars collectively support the effectiveness and reliability of SAGIN, facilitating seamless connectivity and data transmission across diverse environments. In this section, we explore their significance, challenges, and recent advancements and provide a holistic understanding of these SAGIN communication technologies.

In this subsection, we mainly discuss the flexible access methods for the realization of a cohesive and dynamic communication architecture in SAGIN. Within this framework, routing, access, and handover are important issues to build efficient and reliable communication networks.

Flexible access

Routing methods

Routing is of special importance in SAGIN, orchestrating the efficient flow of information across diverse spatial dimensions [23]. The illustration of routing is shown as Fig. 3. In the space and air networks, traditional ground routing paradigms encounter unprecedented challenges due to dynamic trajectories, limited capabilities of computing and processing, and uneven traffic distribution [24]. The integration of satellite, aerial, and ground routers necessitates adaptive routing algorithms to improve the performance of communication pathways. The routing technology has evolved significantly in recent years, driven by the advances in satellite communications, the proliferation of UAVs, and the need for interconnected systems.

It is worth mentioning that in the early stages of satellite systems, routing protocols were primarily based on asynchronous transfer mode (ATM) [25]. However, with the evolution of the Internet and satellite networks, there has been a shift toward advocating for Internet protocol (IP) routing technology. The flexibility and adaptability of IP routing make it a more

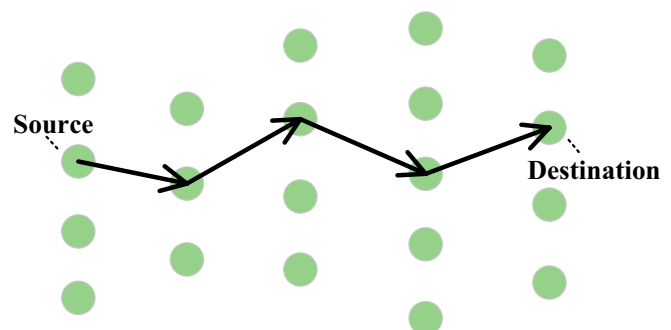


Fig. 3. Illustration of routing.

suitable choice, accommodating evolving network requirements and seamlessly integrating into the global Internet architecture [26].

The primary objective of routing algorithms is to determine the optimal path for data packets from source nodes to destination nodes. This process involves the considerations of network topology, link load, cost, and various other factors. The main categories of routing algorithms include quality of service (QoS)-based routing methods and traffic-based routing methods [6]. The QoS-based routing focuses on selecting the best path based on QoS requirements, including many aspects such as bandwidth, latency, jitter, and reliability. In [27], the authors proposed a routing algorithm with a restoration mechanism, considering constraints such as bandwidth and latency to meet the QoS criteria, and additionally established a restoration topology using a set of network bridges to ensure the provision of lower-cost alternative paths in case of node or link failures. To address potential risks of attacks and disruptions during wireless transmissions, the authors of [28] utilized ant colony optimization to design a perceptual routing algorithm with multiple constraints in vehicular ad hoc networks, providing services that meet QoS, user security, and reliability requirements. A multi-path QoS routing scheme for polar orbit LEO satellite networks was proposed in [29]. Based on a genetic algorithm, this approach achieved lower computational overhead and higher QoS assurance in terms of call blocking probability and delay jitter under the bandwidth and end-to-end latency constraints. In [30], the authors introduced a bidirectional search mechanism combined with a shortest path algorithm to achieve an efficient QoS routing solution under multiple additional constraints. This approach significantly reduced the time for computation, alleviating forwarding state scalability issues, and demonstrated high performance in terms of resource utilization.

On the other hand, traffic-based routing emphasizes optimizing the flow of data in the network to enhance overall performance. This typically involves techniques such as load balancing, congestion avoidance, and traffic engineering. In [31], the authors introduced a multi-layer satellite routing algorithm to enhance the performance of satellite IP networks, particularly demonstrating improved efficiency under high network loads, which effectively reduced communication overhead and computational burden. To address the throughput reduction and increased transmission delays due to the traffic congestion in multi-layer satellite networks, the authors in [4] developed a novel multi-layer satellite network model, which

could alleviate the inter-layer traffic congestion through traffic allocation and significantly decrease packet transmission delays during queuing and propagation. In [32], the authors focused on adaptive routing in the ISL segment of non-GEO satellite systems. Considering 4 different traffic density scenarios, they utilized a simulator to reveal the relative impact of traffic load and propagation delay on the link costs, as well as the effects of traffic factors and model selection on the average packet delay. This work provided the insights of crucial parameters for traffic-adaptive routing, guiding the design and optimization of future satellite communication systems.

Access methods

Access methodologies in SAGIN are also important due to the heterogeneous nature of connected platforms. Satellites, aerial vehicles, and ground stations necessitate specific access methods, ensuring seamless connectivity across the diverse segments of the SAGIN architecture [1]. In contrast to the ground wireless communications, satellite-to-ground transmission is characterized by highly dynamic channel environments, significant variations in user-to-user latency, higher-order Doppler effects, and stringent hardware resource constraints [33]. These distinctive features give rise to signal propagation delays, rapid fluctuations in system parameters, signal structure distortions, and nonlinear channel mappings, presenting systemic challenges to satellite-to-ground communications. Currently, there have been numerous studies in multi-access technology for satellite mobile communication systems.

Satellite access technology has evolved from ground wireless access technology and can be categorized into 3 types: orthogonal multiple access (OMA), nonorthogonal multiple access (NOMA), and random multiple access (RMA) [34] (Fig. 4). In OMA, each user is assigned a unique code, i.e., an orthogonal code, which allows users to transmit in parallel at the same time or frequency without interfering with each other. NOMA technology introduces cleverly designed access feature fingerprints at the transmitting end to ensure controllable interference among users [35]. Advanced multi-user detection algorithms are employed at the receiving end, enabling a significant increase in the number of users accessing the same resources [36]. This makes NOMA a competitive multiple access solution for post-5G/6G scenarios. RMA is a user-unscheduled access method that seizes communication resources in a competitive way, distinguishing itself from the other 2 access methods, and is widely applied in satellite communications. In [37], the authors proposed a NOMA-based irregular repetition slotted

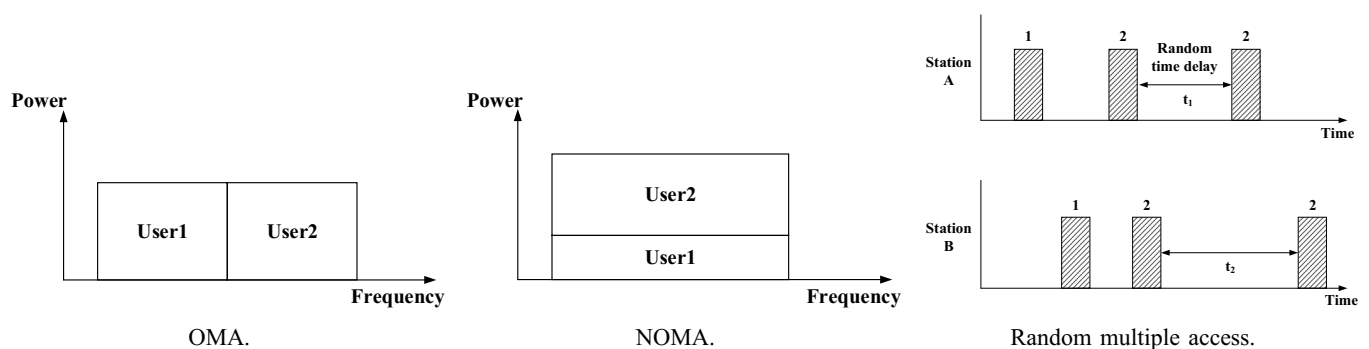


Fig. 4. Access methods.

ALOHA scheme, which mitigated the majority of packet collision issues and effectively reduced the packet loss rates. Simultaneously, they adopted a differential evolution scheme to optimize the throughput of the system, validating the superior performance of the proposed approach over traditional algorithms. In [38], the authors utilized a NOMA satellite link to provide high-speed Internet for aircraft. This was achieved by allowing multiple aircraft to communicate with ground stations simultaneously, thereby increasing system throughput and reducing the required resource time slots. In [39], the authors introduced a new method for random access in satellite networks based on time-division multiple access (TDMA) within the orthogonal slot ALOHA framework. This method incorporated multi-packet detection, enabling higher throughput through the utilization of packet repetition and interference elimination. To accommodate a large number of concurrent machine-to-machine random access requests while ensuring human-to-human communications, the authors in [40] designed an improved random access preamble set and proposed an enhanced random access scheme based on spatial groups, which could significantly reduce collision probability and access latency while achieving robustness against positioning estimation errors.

Additionally, the physical layer technology of multiple access is also of great importance.

In recent years, proposals such as spatial group-based random access [40], frequency-TDMA [41], and NOMA transmission based on irregular ALOHA [37] have provided new applications and perspectives for multiple access. For multiple antenna-based multiple access technology, implementing polarized antennas in massive multiple-input multiple-output (MIMO) point-to-point systems can enhance the system performance and spatial efficiency [42]. Quasi-orthogonal space-time codes can also be applied in dual-polarized MIMO [43].

To deal with the challenges brought by the complex transmission environment changes, numerous studies have been conducted on the end-to-end optimization based on deep learning technologies. To avoid potential computational costs resulting from the direct connection of different artificial intelligence (AI) modules in the training process, investigating neural network-based end-to-end joint optimization methods is necessary [44]. Deep learning can also be combined with massive antennas, and several prototypes further validate the practicality of deep learning at the physical layer [45–47]. Compared with traditional optimization algorithms, deep learning-enhanced resource allocation techniques are more adaptable to complex transmission environments and achieve better performance in communication systems.

Handover methods

The handover technology of SAGIN focuses on the seamless transfer of control and interaction among different communication domains. It is designed to address the challenges of providing uninterrupted communication, which is basically classified into 3 categories: spotbeam handover, satellite handover, and ISL handover [48].

Spotbeam handover involves transitioning a satellite communication link from one spotbeam to another, which is shown in Fig. 5. Spotbeams are focused and narrow satellite beams that are used to cover specific geographic areas. The spotbeam handover aims to maintain connectivity as a device moves across the coverage areas of different spotbeams [49]. It requires

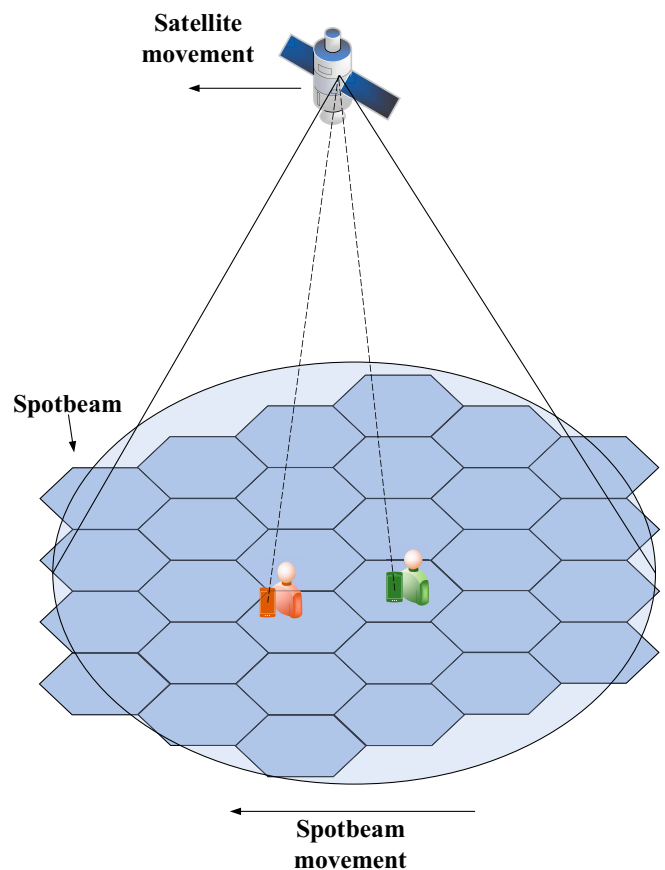


Fig. 5. Spotbeam handover.

the coordination between the satellite and ground systems to smoothly transfer the communication link without significant interruption. Spotbeam handovers are crucial for providing continuous coverage. In [50], the authors proposed a new queuing strategy for the performance evaluation of resource management in LEO mobile satellite systems. Simulations were conducted under 2 technologies, i.e., fixed channel allocation and dynamic channel allocation (DCA), demonstrating the superiority of DCA technology. Additionally, several DCA algorithms were further discussed in [51], revealing the improved performance of DCA in both GEO and LEO satellite scenarios.

While the issues related to spotbeam handover have received thorough investigation, there remains a gap in comprehensive research on satellite handover technologies [52]. This discrepancy arises from the higher frequency of spotbeam handovers compared to satellite handovers. Unlike spotbeam handovers, where users move within a restricted coverage area, satellite handover occurs when a mobile terminal transfers its connection from one satellite to another. This is common in satellite constellations where multiple satellites work together to provide global coverage. As shown in Fig. 6, the handover process involves transferring the communication link from the current satellite serving the user to another one with better signal quality or coverage for improved performance. Satellite handovers are essential for maintaining connectivity in the scenarios where a user moves across different satellites within a constellation. A satellite hybrid handover scheme with channel adaptability and versatility was introduced by [53], and the impact of the handover process on the energy efficiency and service quality

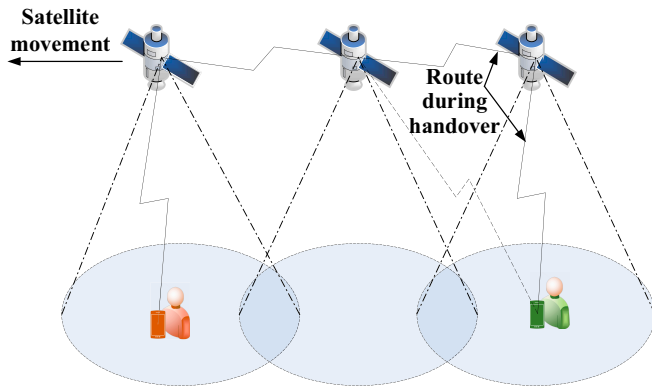


Fig. 6. Satellite handover.

of satellite systems was analyzed. Due to the periodic variation of the LEO network space structure, in order to simplify the network while enhancing user service provision, a new handover technique for satellite fixed cell system was proposed in [54] to efficiently utilize system resources and provide an equable handover in satellite systems.

Furthermore, when examining the intricacies of satellite handover, it becomes evident that seamless communication involves not only transitions between satellites but also efficient management of ISL. The satellite handover primarily focuses on user mobility across different satellites, while ISL handover addresses the handover process between interconnected satellites, emphasizing the importance of optimizing data traffic routing among satellites.

In a satellite constellation, multiple satellites are interconnected through ISLs to facilitate communication handovers and improve overall system performance [55]. ISL handover involves transferring data traffic from one ISL to another, optimizing the routing path based on the factors such as signal strength and available resources (Fig. 7). This technique ensures efficient communication between the satellite constellations, contributing to seamless connectivity and load balancing. In [56], the authors presented a design for an ATM-based ISL handover scheme in LEO and MEO satellites. The proposal involved enhancing the shortest path algorithm to accommodate the dynamic ISL network topology, significantly reducing the network latency jitter, and simplifying the alignment of ATM cell flows. In [57], a pseudo-random scheduling architecture based on synchronous spatiotemporal partitioning was employed to adapt to the high dynamic environments in ISL, offering strong flexibility and robustness while significantly reducing power consumption and bandwidth requirements.

Efficient resource scheduling

As SAGIN involves a variety of resources from space, air, and ground, the network topology is more complex, which often results in inefficient resource utilization and influences communication quality. Therefore, it is essential to develop efficient resource scheduling methods that can enhance resource utilization to improve the network performance. In the following, we will explore efficient resource scheduling methods for SAGIN.

Time-frequency resource scheduling

In the initial stages of satellite communications, the system mainly adopted methods, such as frequency-division multiple

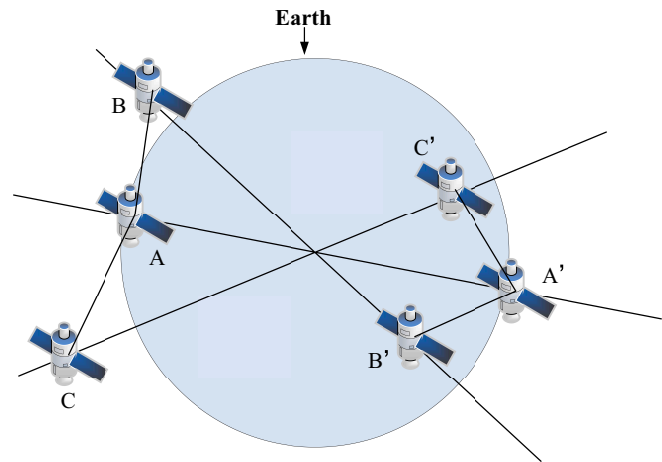


Fig. 7. ISL handover.

access (FDMA) and TDMA. In FDMA, the available frequency band is divided into multiple nonoverlapping subchannels, and each user is assigned a specific subchannel for their exclusive use. TDMA technology divides communication resources into time slots and assigns different time slots to different users. The time-frequency resource scheduling issue is mainly investigated in the communication mechanism of multi-frequency time-division multiple access (MF-TDMA). The MF-TDMA scheme combines the advantages of FDMA and TDMA techniques, and is favored by many modern satellite communication systems [58]. The MF-TDMA technology first divides the channel resources into a series of different frequency carrier channels based on FDMA mechanism, and the TDMA technology is then used to further divide each carrier channel into multiple time slots, as shown in Fig. 8.

Traditional MF-TDMA satellite communication systems employ static slot and frequency allocation methods, which are difficult to dynamically adapt to real-time demands. Dynamical time-frequency scheduling systems can dynamically determine the allocation of time slots and frequency resources based on certain algorithms and protocols according to the characteristics of users and channels. When a user requests communications, the resources can be allocated in 2 steps. First, select an appropriate subcarrier channel based on the type of services and hardware capabilities of terminals. Then, calculate the required time slots based on the link conditions of the selected subchannels and resource allocation constraints. Dynamical time-frequency resource scheduling allows for more flexible resource allocation and improves the system capacity. In [58], the authors employed a Markov model-based prediction scheme to enhance the accuracy of resource calculation-resource assignment phase incorporated a novel packing algorithm for efficient time slot allocation. In [59], an efficient algorithm was proposed to determine the super-frame design that maximized the system throughput in the resource calculation phase, and a novel hybrid bin-packing algorithm was introduced to maximize the utilization of the MF-TDMA frame.

Beam hopping

In space networks, multi-beam satellites play a crucial role in efficiently utilizing limited resources to meet the requirements of communication. In a traditional multi-beam satellite system,

satellite communication resources are preassigned to different beams, with all beams illuminated simultaneously and unable to change direction [60]. However, due to the uneven geographic distributions and rapid changes in the demands of ground nodes, the efficiency of beam resource utilization is relatively low, and the spatial freedom of the beams is not fully exploited. Therefore, beam hopping technology is expected as

a solution to address the above issues in multi-beam satellite systems, which is depicted in Fig. 9.

Beam hopping is based on time slicing technology, utilizing fewer beams to cover more cells. At different time intervals, the beams hop among different cells. Beam hopping achieves flexible resource allocation and signal coverage through purposeful activation or deactivation of beams, and it can also reduce payload,

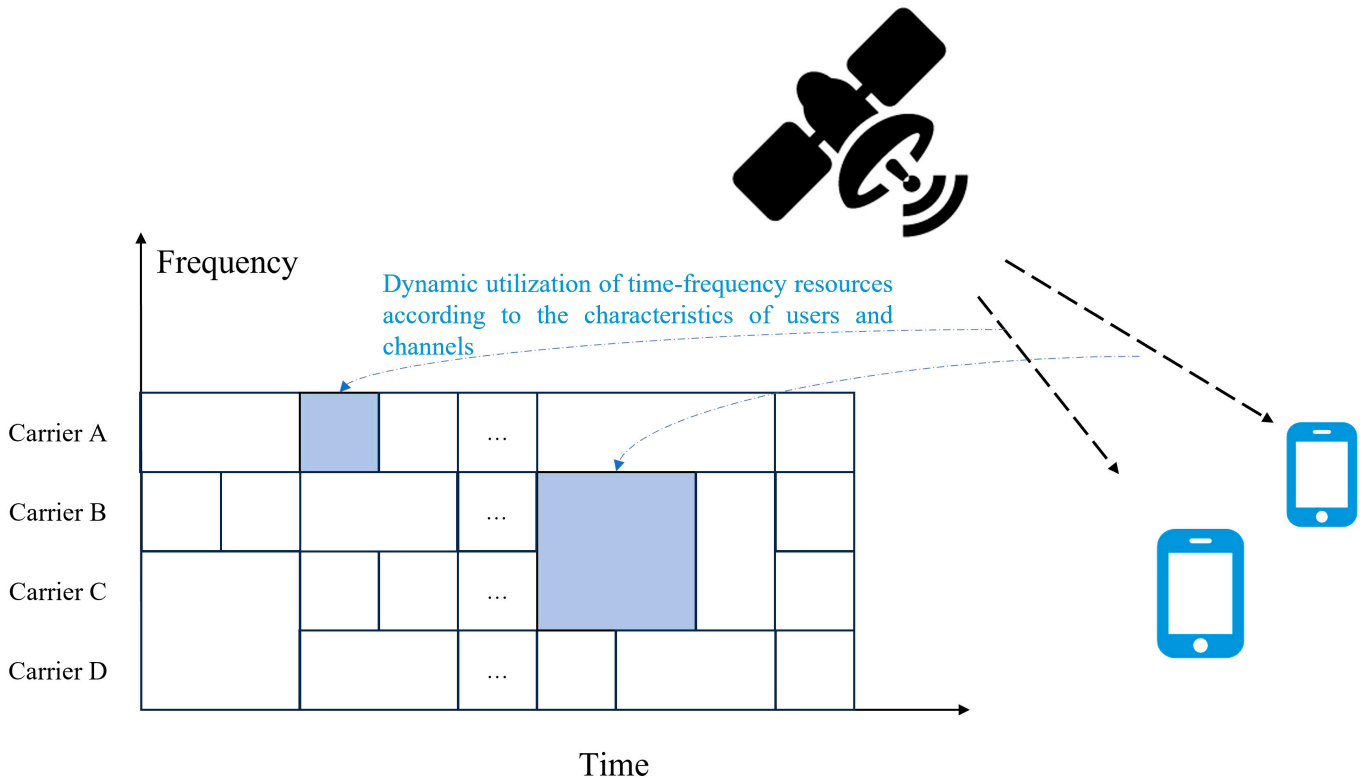


Fig. 8. Dynamical MF-TDMA.

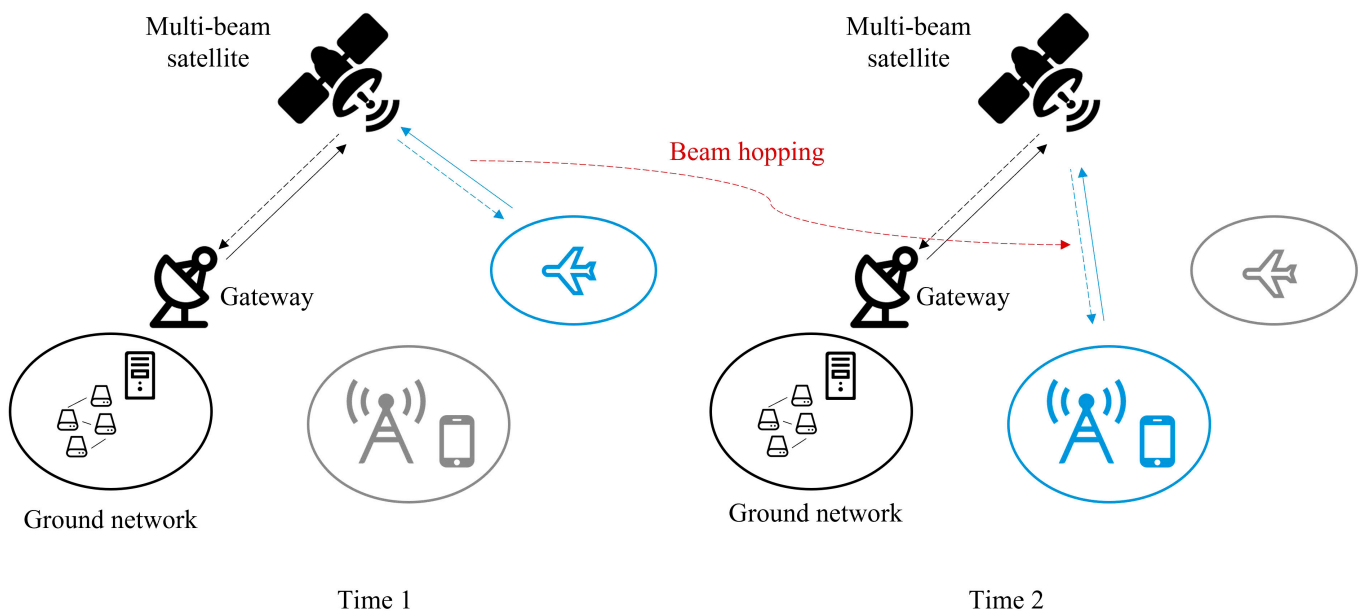


Fig. 9. Beam hopping.

thereby lowering costs. Therefore, beam hopping is a flexible solution to address the rapid increase and uneven distribution of data traffic. In addition, the method of activating beams at greater distance apart can be employed to avoid co-channel interference caused by spatial factors [61].

Recently, there has been extensive research on the design of beam hopping schemes. In [62], a wireless resource allocation scheme was developed for DVB-S2X to exploit the beam hopping across cells, which could ensure the fairness among beam services and minimize the real-time business transmission latency while maximizing the throughput for non-real-time business transmission. For multi-satellite beam hopping, the authors in [63] introduced a method based on load balancing and interference avoidance. This method could allocate cells to different satellites to balance their traffic load and select beam hopping patterns that meet the spatial isolation condition to avoid intra-satellite and inter-satellite co-frequency interference. Furthermore, slot scheduling was employed to meet the traffic demand of each cell as much as possible. The authors in [64] decomposed the multi-satellite joint beam hopping into satellite-cell assignment subproblem and joint multi-satellite beam hopping pattern design subproblem. In [60], the authors considered the joint optimization of beam hopping and NOMA by adjusting power allocation, beam scheduling, and terminal-time slot assignment. In [65], the authors fully exploited the degrees of freedom in terms of time, frequency, and space, and introduced a dynamic beam pattern and bandwidth allocation scheme based on deep reinforcement learning.

Furthermore, the European Space Agency proposed and studied the cluster hopping scheme, dividing the entire virtual multi-beam grid into a set of fixed-sized and fixed-shaped nonoverlapping clusters [66–68]. In [61], the authors developed a general framework of dynamic and flexible cluster hopping, allowing for the activation of adjacent beams simultaneously based on request where linear pre-coding was utilized to eliminate intra-cluster interference.

UAV trajectory

As an aerial communication platform, UAVs can be flexibly deployed as required, which are commonly used as aerial base stations or aerial mobile relays, offering communication services for the whole wireless infrastructure [69,70]. Compared with space-based satellite communication networks and dense ground-based communication networks, the cost of UAV communications is relatively low [71]. The deployment of UAVs can be categorized into static deployment and dynamic deployment. In the static deployment, the UAVs are acted as fixed aerial base stations at a fixed altitude to offer network support. The research on UAV static deployment primarily focus on the UAV deployment/placement optimization to meet the coverage requirement of a given set of ground users with the minimal UAV base stations required [72]. In the dynamic deployment, the location of the UAV changes instantaneously, and the energy consumption of UAV networks is mainly attributed to the energy consumption during flight and communication, which is closely related to the deployment of the UAV [73], as shown in Fig. 10. In summary, the trajectory of the UAV can influence the system capacity and power consumption of wireless communications [74,75].

Currently, the UAV trajectory optimization is of special importance to enhance the efficiency of multi-UAVs collaborative and energy consumption optimization. In [76],

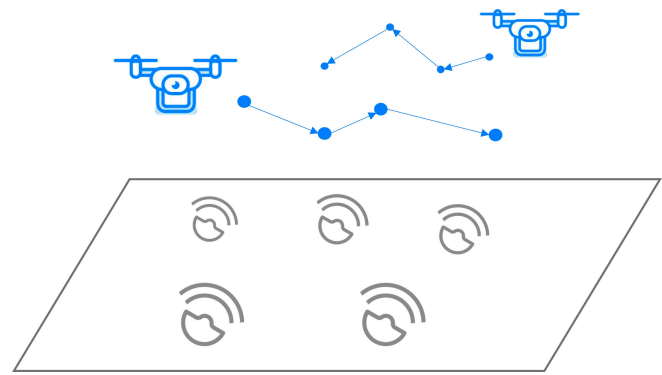


Fig. 10. UAV trajectory.

the authors provided a general definition of the efficiency of UAV communications and investigated the energy efficiency of rate maximization and energy minimization for the case of unconstrained UAV trajectory, and the energy efficiency maximization for the case of the general constraints on UAV trajectory, including its initial/final locations and velocities, minimum/maximum speed, and maximum acceleration. An efficient algorithm based on linear state-space approximation and sequential convex optimization techniques was proposed to find the approximate optimal trajectory. The authors in [77] jointly optimized the UAV trajectory and the transmission power allocation of the source and the UAV relay, through block coordinate ascent and successive convex approximation methods, to maximize the throughput. For multi-UAVs that enabled wireless networks, the authors in [72] jointly optimized the user scheduling and association, trajectories, and transmitted power to maximize the minimum average rate among all users. In [78], the authors formulated multi-UAV trajectory optimization as a sequential decision-making issue and proposed a decentralized deep reinforcement learning algorithm. With the rapid development of machine learning, how to use machine learning to design the trajectories of multiple UAVs is worth further investigation.

Spectrum sharing

In space-ground networks, the issue of spectrum scarcity is particularly serious, leading to increased competition for spectrum resources with the existing ground networks [79]. The growing scarcity of spectrum resources and limited efficiency of current spectrum allocation strategies have posed obstacles to the development of space-ground networks [80]. To alleviate the pressure on limited spectrum resources and maximize the network utility, spectrum sharing has been introduced in the space-ground networks. Spectrum sharing includes static spectrum sharing and dynamic spectrum sharing. Specially, in the static spectrum sharing, the spectrum allocation is done only once before starting the transmission of secondary users, and no hand-off is performed during the transmission [81]. In the dynamic spectrum sharing, the spectrum resources are dynamically allocated based on the real-time demand, which offers greater flexibility than the static spectrum sharing and is regarded as a viable solution to tackle spectrum scarcity issue. Cognitive radio, which has the capability to sense and utilize idle frequency bands and enables dynamic spectrum sharing, is an important technology for future wireless communication networks [82]. In the conventional cognitive

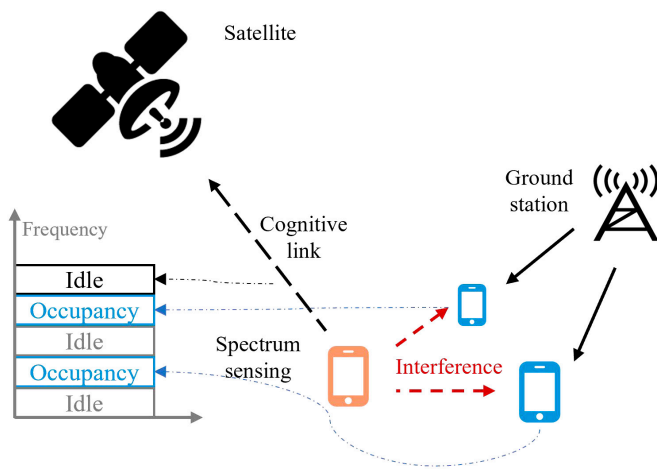


Fig. 11. Spectrum sensing.

radio situation, a secondary user is permitted to utilize the spectrum that is held by the primary user only at the time when the primary user is detected to be idle or if any potential interference to the primary user can be sufficiently controlled [83].

Spectrum sensing is an essential component of cognitive radio. It serves as a prerequisite for achieving dynamic spectrum sharing, especially in space-ground networks. Secondary users can detect the spectrum occupancy states with a high probability and accurately determine whether to access a specific frequency band by adopting spectrum detection strategies, which avoids network congestion and mutual interference [84], as shown in Fig. 11. The cognitive radio technology based on spectrum sensing has been extensively studied [85,86], and a series of spectrum sensing algorithms have been developed, such as matched filter detection [87], energy detection [88], and eigenvalue-based detection [89]. Considering the long propagation distance of SAGIN, spectrum sensing is required to be done in low signal-noise ratio (SNR) conditions [90]. In such scenarios, the reliability of traditional spectrum sensing methods will decrease, and a deep learning-based spectrum sensing scheme was developed in [90], the detection performance of which was improved in a low SNR regime. In [91], the authors considered a sensing-based cognitive space-ground network that combined a cognitive space-ground network with distributed cooperative spectrum sensing network to maximize the energy efficiency.

Secure communications

In SAGIN, the network topology is highly unstable due to the dynamic movement of mobile nodes such as satellites and UAVs, making space-ground and aerospace channels particularly susceptible to malicious attacks such as eavesdropping, jamming, and unauthorized access. Hence, the security is one of the most critical issues for wireless communications in SAGIN, and ensuring the confidentiality, integrity, and availability of transmitted data is of special importance. The primary objective of secure communications is to ensure that the receivers can get the transmitted data without any jamming or unauthorized access while preventing any eavesdroppers or wiretappers from deciphering or understanding this information [92]. In this subsection, we mainly discuss covert communication, physical layer security, and anti-jamming communication, as shown in Fig. 12.

Covert communications

Traditional secure communication technologies focus on protecting the data content but cannot hide the wireless transmission behaviors. However, the leakage of transmission behaviors can lead to the exposure of confidential tasks. Therefore, in the scenarios where security and privacy requirements are stringent, not only does the content of communication need to be kept confidential, but also the process of data transmission requires to be hidden. In order to provide more direct and effective protection for the security and privacy of SAGIN, covert communications have emerged as a new paradigm for wireless security, attracting significant research interest in recent years [93].

Covert communications, also known as low probability detection communications, aim to utilize the uncertainty in wireless channels to hide the transmission process and reduce the probability of being detected by third parties while maintaining reliable communications at a certain rate. Direct sequence spread spectrum (DSSS) is the most classical and commonly used covert communication technique, where the signals are hidden within noise by spreading the signal across a wider spectrum. The DSSS can efficiently reduce the susceptibility to interception, thereby achieving hiding and protection of signals [94]. In [95], the authors derived the theoretical limit of information-theoretic for covert communications in additive white Gaussian noise (AWGN) channels, which is also known as the square root law (SRL). Under surveillance, it is possible to reliably and covertly transmit a maximum of $O(\sqrt{n})$ bits simultaneously to the intended receiver across n channels. The SRL indicates that when n is sufficiently large, the rate of covert communication becomes zero, i.e., $\lim_{n \rightarrow \infty} \sqrt{n}/n = 0$.

The source of covertness is the uncertainty of warden's detection [96]. In order to obtain a positive covert rate, a series of techniques utilizing noise and channel uncertainty were proposed to generate more uncertainty [97]. In [98], the authors demonstrated that artificial noise (AN) could significantly improve the communication rate of covert communications. However, the continuous transmission of AN signals by the jammer required more additional energy. In [99], the authors proposed a method of probabilistically transmitting AN with variable power by the jammer against detection and enhanced the energy efficiency of the system. In [93], the authors explored the potential of cooperative jamming in covert communications and proposed a scheme for uncoordinated jammer selection. Furthermore, the authors in [100] investigated the covert communications utilizing a cognitive jammer, which demonstrated that the cognitive capabilities of the jammer could enhance the communication rate while effectively preventing the eavesdropper from detecting the transmission. Furthermore, in [101], different from the prior covert backscatter works using AN, the authors achieved covertness by arming the backscatter tag with multiple antennas.

Recently, the concept of intelligent reflecting surface (IRS) has been proposed as an enabling technology to customize wireless propagation channels that can improve the quality of received signals [102]. The adoption of IRS can further benefit from the uncertainties brought by channel fading [103]. IRS can change the characteristics of reflected electromagnetic waves, such as phase, polarization, and direction. By integrating IRS into the system, the transmission channel quality can be enhanced

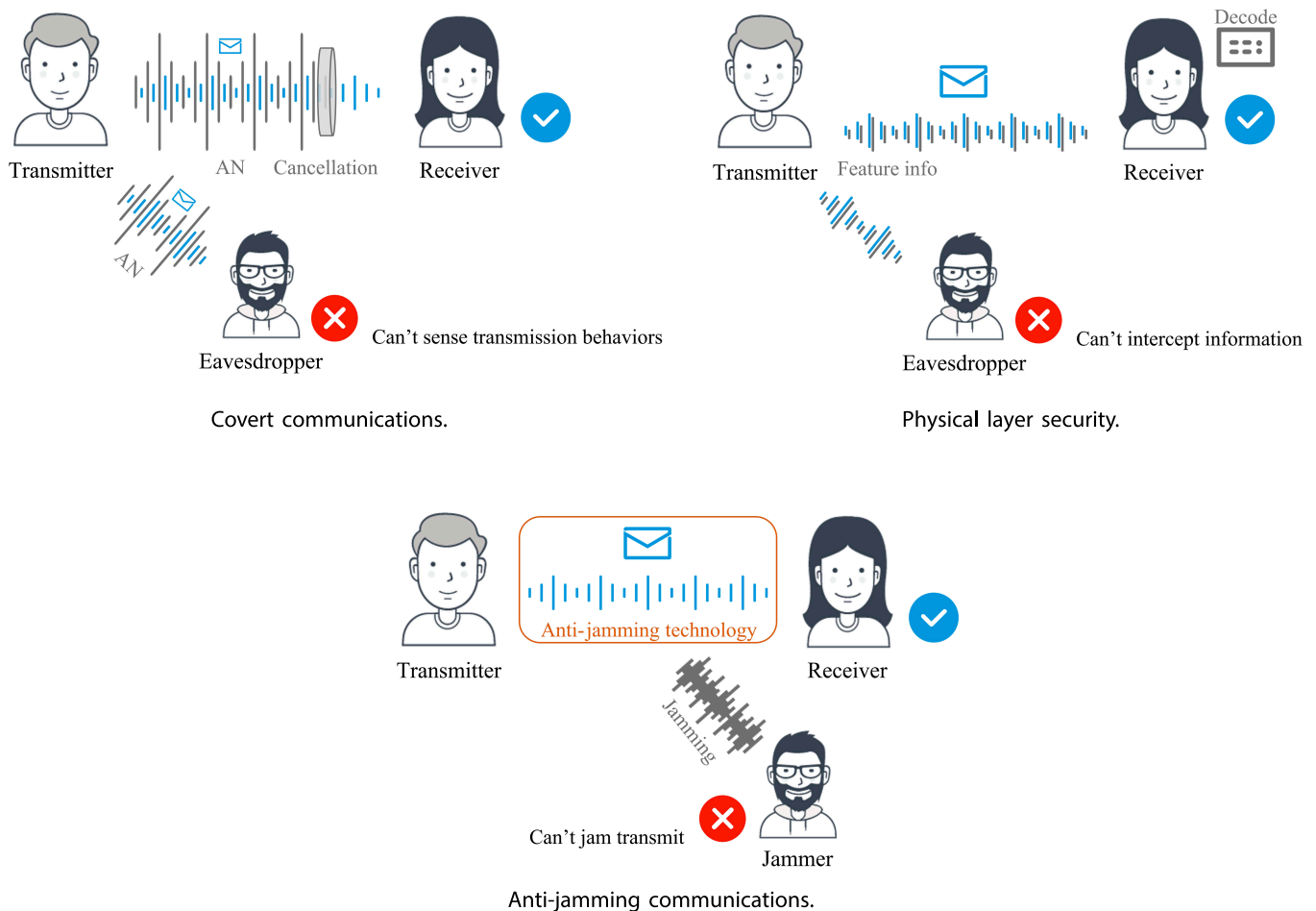


Fig. 12. Secure communications.

without the need to increase transmitter power. It not only leads to lower overall power consumption but also effectively improves the performance of steganographic of the system [104].

Physical layer security

Physical layer security is a method that utilizes the characteristics and vulnerabilities of the propagation medium to ensure the secure transmission of information [105]. Different from covert communications, which aim to hide communication behaviors from the eavesdropper, physical layer security attempts to minimize the interception of information by eavesdroppers [106], showcasing advantages of lower complexity and resource savings. From the perspective of information theory, it has been demonstrated that physical layer security can achieve perfect secrecy, even in the presence of eavesdroppers with significant computing capabilities [107]. Physical layer security can be traced back to the eavesdropping channel model, which was proposed by Wyner in 1975 [108]. It utilizes the differences between legitimate and illegitimate channels to achieve confidentiality at the physical layer without the need for any key. Currently, physical layer security technologies can be categorized into the following 3 types: physical key generation, physical coding, and physical layer authentication [109].

Due to the difficulties in key distribution and excessive reliance on mathematical complexity [110], the traditional cryptographic mechanisms are not suitable for SAGIN. Unlike

conventional cryptographic mechanisms, physical key generation focuses on the physical layer channel. It utilizes the entropy of randomness in the transmit–receive channels, such as channel state information and received signal strength, to generate secrecy keys for communications, thereby safeguarding the data information that needs to be transmitted at the physical layer [109]. Physical coding serves as a foundational process in communication technology, aiming to maximize the transmission rate of data among authorized users. Note that improving physical coding can also effectively mitigate attacks on data integrity, such as unauthorized modifications occurring during message transmission. Currently, the commonly used coding methods are low-density parity check and polar codes, which have shown excellent performance in terms of physical layer security [111].

Physical layer authentication is one of the core technologies to ensure the security of wireless communications. The basic principle of this technology is to combine the physical communication link with the spatial–temporal specificity of the transmitted signals to authenticate the physical channel characteristics of the communicating parties, and identity authentication is achieved at the physical layer [112]. It has the advantages of fast authentication speed, low complexity, and compatibility, without the need to consider various protocol implementations. Based on different authentication criteria, the mainly investigated authentication techniques can be

classified into 3 categories: radio frequency (RF) fingerprint authentication [113], physical layer signal water-making authentication [114], and physical layer wireless channel authentication [115]. Among them, RF fingerprint authentication was achieved through the inherent random hardware impairments of amplifiers and oscillators in wireless devices [116]. The presence of random hardware impairments introduced distinctive deviations in RF signals within the time–frequency domain, including variations in amplitude, frequency, and phase [117]. Physical layer watermark authentication is a technique that involves transmitting both the key and the message simultaneously while concealing the authentication information within the signal. This approach ensures that the presence of the authentication signal is not easily detectable, and it does not require any additional bandwidth. Physical layer wireless channel authentication is a technique that leverages the distinctiveness of channel state information between different transmitters and receivers to authenticate user identities. Common channel characteristics, such as channel impulse response and received signal strength [118], can vary with changes in users' spatial location. These features serve as identity detection metrics for different users in various scenarios.

Anti-jamming communications

Jamming is caused by deliberate emission of interference signals from jammers to disturb wireless communication networks, which results in throughput degradation, network partition, or even a complete zero connectivity scenario [119]. In addition, malicious jammers have the ability to intercept and manipulate transmitted data, as well as impersonate sensor nodes to transmit inaccurate information [120]. Different from other security challenges, such as eavesdropping and data fabrication, wireless networks exhibit much more serious susceptibility to radio jamming attacks. Especially, satellites are vulnerable to the malicious jamming attacks because of the cyclic visibility and fixed orbit [121], and the communications among the satellites and UAVs in the open air are also challenged by the jamming, which is dominated by the line-of-sight transmissions [122]. Nowadays, the technologies for conducting jamming attacks are simple and the equipments are relatively inexpensive [123], and jamming has been a serious threat in SAGIN.

Wireless signals have characteristics such as time, frequency, power, and space, and if the jamming overlaps with the useful signals in these characteristics, the communication will be affected by jamming. The issue of anti-jamming is to make the valid signal avoid overlapping with the jamming as much as possible. Even if there is an overlap, the signal processing techniques should be used to ensure that the output of the receiver has a higher signal-to-interference power ratio, known as SIR. Therefore, effective anti-jamming technologies involve processing signals in terms of time, frequency, power, and space domains to improve SIR. For instance, typical anti-jamming technologies in time domain include time hopping [124] and adaptive error control. In the frequency domain, it is common to utilize frequency hopping spread spectrum [125], adaptive frequency control, or frequency diversity [126] as an anti-jamming measure. In the power domain, anti-jamming objectives can be achieved by suppressing the interference power while maintaining or increasing the signal power, or enhancing the signal power when alteration of the interference power is not feasible. Boss DSSS [127] and adaptive power control can be classified as power domain technologies. Besides, there are

some technologies considered from the space domain, such as adaptive array processing [128] and beamforming [122], to optimize the utilization of spatial characteristics for enhancing the desired signals and suppressing the interference.

However, with the network topology and electromagnetic environment becoming more complex, as well as the development of jamming technology, it is very difficult for traditional single-domain anti-jamming schemes to satisfy the reliability requirements of wireless communications. The diversity, dynamics, and intelligence of jammers pose more strict demands on anti-jamming technologies [129]. In [130], a multi-carrier DSSS method that combined time-domain and frequency-domain approaches to mitigate narrowband jamming was developed. In [119], the authors explored the use of MIMO technology and orthogonal frequency-division multiplexing that combined frequency domain and space domain to achieve anti-jamming communications. The authors in [131] exploited spread spectrum and user mobility to address jamming. Moreover, the authors in [132] presented a novel active anti-jamming scheme based on jamming modulation, which could transmit data over jamming signals, instead of mitigating them as traditional schemes. Furthermore, orbital angular momentum provided a new mode domain, which was the mode multiplexing technique in the field of vortex-electromagnetic beams. In [133], the authors jointly used mode hopping and traditional frequency hopping to further decrease the bit error rate for wireless communication.

As a further development, AI technology provides jammers with more diverse and intelligent jamming attacks, and jamming intelligence is an irresistible development trend of electronic jamming attacks [134]. In [129], a joint multi-agent anti-jamming algorithm was proposed to obtain the optimal anti-jamming strategy. The authors in [121,134] proposed 2 distinct jamming mitigation solutions based on game and reinforcement learning, respectively. Furthermore, the authors in [135] considered the use of an IRS to enhance anti-jamming communication performance against a smart jammer, which could jointly optimize power allocation and reflected beamforming at the IRS.

Future Directions

In this section, we briefly introduce some potential research directions for SAGIN.

Edge computing and intelligent offloading

Edge computing and intelligent offloading are very important for satisfying the network requirements of future applications that have large computational tasks or ultra-low latency needs, which are also of significance in the seamless communication and real-time computation of SAGIN [136]. In SAGIN, an abundance of sensors, detection devices, and data acquired via satellites or other sources are involved. The combination of edge computing and intelligent offloading offers the capability for rapid data processing near the data sources, facilitating the swift monitoring and timely response to unforeseen events. In recent years, aerial vehicles used in edge computing, such as unmanned drones and balloons for data transmission or task execution, have received lots of attention [137]. Edge computing and intelligent offloading can be employed for performing real-time image processing, obstacle recognition, trajectory planning, and other tasks on aerial platforms, thereby augmenting the

autonomy and adaptability of the aircraft. Simultaneously, implementing intelligent offloading methods about the QoS and quality of experience for SAGIN can contribute to the effective resource scheduling of SAGIN as well as balancing the overall distribution of task workloads [1,138].

Deep space communications

Delving into the depth of the solar system and the cosmos through deep space exploration offers the probability of discovering potential resources or signs of life that may exist on other planets and celestial bodies, which is of profound significance. In this scenario, deep space communications become an indispensable and pivotal element for the success of such explorations. Starting from the moment of liftoff, the sole link between a spacecraft and the earth is established through the communication system, which facilitates the data transmission from spacecraft back to the earth, tracks the spacecraft, and issues commands for actions that the spacecraft cannot autonomously perform [139]. Presently, several nations, including the United States, Russia, China, European Union, India, and Japan, have actively participated in deep space exploration and have established deep space networks [140]. However, considering the constraints of the payload power, mass, and antenna size of the satellites, along with the features of deep space environment, which include high latency, narrow bandwidth, and high error rates, conducting long-term research on the key technologies and detection methods for deep space communications is of vital importance [141].

Satellite-based distributed machine learning

Although current research has equipped satellite networks with edge cloud servers to provide computational offloading for remote devices, bringing processing heavy applications to remote areas and ensuring ubiquitous connectivity in future 6G networks, the analysis of the significant data generated by these remote devices remains a challenge [142]. Under the circumstance, satellite-based distributed machine learning can leverage satellite networks for distributed data processing and model training, connecting various spots on the earth through satellite systems, to enable collaborative processing of large-scale data for real-time decision-making analysis. Particularly, utilizing satellite networks enables the integration of dispersed data from around the globe into a unified system, which is critical for investigating and addressing the global issues, such as climate change monitoring and natural disaster prediction [143]. In summary, investigating satellite-based distributed machine learning has the potential for SAGIN to enhance the timely response to emergencies and predictive maintenance applications, and additionally, distributed learning contributes to safeguarding user privacy during data transmission, making it of paramount significance.

Blockchain-based SAGIN

As an emerging distributed ledger technology, blockchain is meant to achieve the security, transparency, and traceability of information, through the way of sharing and maintaining an ever-growing and tamper-proof data chain among multiple nodes in the network [144]. The security attributes inherent to blockchain, including immutability, trusted traceability, decentralization, distrust, cryptographic algorithmic security, and nonrepudiation, empower its capacity to deliver a high level of

security, transparency, and efficiency in data management and interactions in SAGIN, which in turn promotes the development of digitization and intelligence [145]. For instance, considering the unique architecture of SAGIN, a substantial amount of sensitive data and communications may be involved, while there potentially exists the risk of single point of failure in traditional security service schemes, since they often rely on a trusted third-party central node. In contrast, blockchain-based security services guarantee the security of data by leveraging cryptography and decentralized features, reducing the risk of single-point failure as well as enhancing the robustness and stability of SAGIN [146]. In summary, blockchain-based security services represent a promising paradigm for addressing data security issues in SAGIN, and research on the integration of blockchain with some emerging technologies, such as IoT, AI, and edge computing, will contribute to SAGIN.

Quantum communication integrated SAGIN

Quantum communication is based on the principle of quantum mechanics for information transmission [147]. Since it is impossible to achieve an accurate replication of an unknown quantum state in quantum mechanics, and any eavesdropping behavior will be detected by both sides of the communication, quantum state transmission can fundamentally ensure the “unconditional security” of communication [148]. The integration of quantum communication into SAGIN can not only improve the security of network communications but also expand the coverage and enhance the network performance. Specifically, first, the cryptographic keys provided by quantum key distribution can be used to protect the transmitted data and enhance the security of SAGIN, which is critical for the transmission of the military and sensitive information [149]. Second, quantum communication through satellites can overcome the distance limitation of terrestrial optical fiber transmission and achieve secure communications on a global scale. Besides, using HAPs as relays, quantum communication networks can be rapidly deployed at disaster sites or in areas where high-density communications are temporarily required, further expanding and enhancing the coverage of terrestrial communication networks. The coverage of ground base stations can be extended by quantum satellite relays. Third, quantum repeater and entanglement swapping technology can extend the effective distance of quantum communications, solve the attenuation problem of quantum signals in long-distance transmission, and make it possible for SAGIN to achieve long-distance, high-speed, and secure data transmission [150]. In summary, integrated quantum communications will help SAGIN better support the diversified applications, such as military communications, financial transactions, scientific research, and government communications, and promote the development and maturity of the related quantum communication technologies, such as efficient quantum detectors and stable quantum repeaters.

Conclusion

In this overview, we have discussed the background and benefits, basic architecture, and development prospects of SAGIN. Then, we have presented some main challenges faced by SAGIN from the perspective of network architecture, node topology, communication channels, and resources. After that, the key technologies in SAGIN from the perspective of flexible access, efficient resource scheduling, and secure communications have

been discussed. Finally, some future directions of SAGIN have been explored.

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Data Availability

There are no data associated with this article.

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Communications in Space–Air–Ground Integrated Networks: An Overview

Kai Yang, Yichen Wang, Xiaozheng Gao, Chenrui Shi, Yuting Huang, Hang Yuan, and Minwei Shi

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As an emerging vertical heterogeneous network architecture that integrates the satellite-based space networks, air networks, and traditional ground networks, space–air–ground integrated network (SAGIN) is developed to realize the Internet of everything, global coverage, and ubiquitous intelligent communications. However, SAGIN also faces quite a few challenges due to its unique characteristics, such as highly complex network architecture, highly dynamic node topology, time-varying communication channels, and restricted resources. In this paper, we first introduce the architecture and benefits of SAGIN, and then present the faced challenges. Next, we discuss some key technologies in SAGIN. In particular, we discuss the flexible access in the realization of a dynamic communication architecture, explore some efficient resource scheduling methods to avoid wastage of resources, and investigate the secure communications, such as covert communications, physical layer security, and anti-jamming communications, in SAGIN. Finally, the potential future directions are discussed.

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